E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23614955

RECEIVED 23.04.2026

ACCEPTED 11.05.2026

PUBLISHED 30-06-2026

Salicylic Acid in Psoriasis Management: Keratolytic and Therapeutic Implication- A Review

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ABSTRACT:

Psoriasis is a chronic skin disease that is caused due to hyperproliferation of keratinocytes (uncontrolled skin cell development), this is generally caused by various factors such as genetics, environment, lifestyle etc. Unfortunately, this disease cannot be treated permanently, it might occur after the treatment. so, management of psoriasis is essential to reduce the chances for worsening the psoriasis condition. Salicylic acid is a keratolytic agent that helps to remove dead cells from the applied area or site and cleans the pores, helping in the psoriasis treatment and management options. Salicylic acid enhances the activity of topical treatment options available for the psoriasis such as steroids, turmeric, coal tar etc. Salicylic acid is effective, safe, affordable. Researcher's articles are used to analyze salicylic efficiency in the psoriasis management and treatment.

Keywords: Psoriasis, Salicylic acid, Keratolytic agent, Salicylic acid in psoriasis

1.Introduction

Psoriasis

Psoriasis is a common skin disorder that occurs on skin due to various reasons (Kasolang et al., 2020). Psoriasis is a life-long immune mediated skin related to inflammatory disease; it also drives some more problems with it such as psoriatic arthropathy, psychological, cardiovascular and hepatic diseases. WHO (world health organization) described psoriasis as a non-infectious disease and highlighted as psoriasis is related to stress and wrong diagnosis, improper treatment and labeling of this disease (Michalek et al., 2016). Psoriasis doesn't pass person to person; it is not a contagious disease and doesn't have any proper treatment yet. Red itchy scaly patches commonly occur on knees, elbows and skin indicates the symptoms of psoriasis(George et al., 2017).psoriasis influences 2-3% population of the of the world (George et al., 2017).

Diagnosis

Psoriasis is generally diagnosed by doctors and clinical practitioners by their appearance (Papp et al., 2011).

Table 1. Symptoms of psoriasis

Disease	Observation
Plaque psoriasis	• **Properly defined, redness, dry scaly patches bigger than 0.5 cm, all over the body or at one site.** • **Differentiated as pe the site of disease**
• **Flexural**	• **Its other names are intertriginous or inverse psoriasis** • **Properly defined, improper scales, patches limited at body folds (under the breast, underarms, genital area etc.)**
• **Nail**	• **This could also occur without skin plaques** • **Small pitting, nail separation from tip, accumulation of dead cell at bottom, oily patches, bleeding in nails, white spots in nails, broken nails, red spots at tip and bottom of nail bed** • **Patients with nail psoriasis are highly prone to psoriatic arthritis**
• **Scalp**	• **Highly prone area for psoriasis** • **Hard to treat**
• **Palmoplantar**	• **Usually occur and limited to hands palm and foot soles** • **Continuous redness and scaling without plaques, and can be turned to poor scaly, fissured areas or bigger plaques spread to palm or foot sole**
Different types	**Observation**

• Guttate	• Small acute, lite pinkish papules like tiny drop at body or limbs` • Symptoms can occur after group A streptococcus infection
• Erythroderma	• Slow developing or acute redness covers 90% or more of patient body with less scaling • Linked with hypothermia, low albumin in blood, instable ions, heart failure • Chronic and dangerous for life
• Pustular	• Inflamed skin with painful pustules, filled with pus forms into patches • Usually occur and limited at palms or soles
• Annular	• Well defined, red scaly patches having clear center
This information is takin from different articles (Papp et al., 2011)(Kim et al., 2017)**.**	

Salicylic Acid

Salicylic acid is a beta acid (BHA), which is classified under the phenolic acid and monohydroxybenzoic acid category. Metabolized salicin forms salicyclic acid. Its chemical structure is similar as aspirin (acetylsalicylic acid) but commonly it is used for acne treatment. Salicylates are the easters and salt of the salicylic acid, chemically these are part of plant phenolics having an aromatic ring and hydroxyl group and physically they are crystalline powder melt at 157–159°C, hydrophobic and lipophilic nature(Raskin, 1992). Aspirin immerge to water-based solution shows hydrolysis reaction and breaks into acetic acid and salicylic acid (Mitchell & Broadhead, 1967). 36 different categories of plant was used to identify the salicyclic acid (RASKIN et al., 1990).Rice, barley, crabgrass and soyabean plants have salicylic acid around 1mg/g fresh mass. The amount of salicylic in 27 thermogenic plants leaves and 7 plants flower found different, but in flowers of some thermogenic plants such as palmae family, bactris major jacq, water lily flowers have no salicylic acid(RASKIN et al., 1990).

properties of Salicylic acids

SA is one of the secondary metabolites, “Secondary” indicates that they have little use or on significant use of it in the plants. But it was noted that some phenolic compounds are essential for plant growth, development, and for other organisms health and development (Janzen, 1981).

Free salicylic acid (SA), a crystalline powder melts at 157–159°C, these are highly soluble in polar organic solvents and moderately soluble in water. The pH value of saturated aqueous of salicylic acid is 2.4. when it is saturated in 301 nm, SA florescence at 412 nm; this makes it easier to differentiate this molecule in the variety of plant system (Raskin et al., 1987)(RASKIN et al., 1990). The HPLC method is used to extract, analyze and identify the salicylic acid present in the plant was modified form (Raskin et al., 1989) appears in

Yalpani et al (Yalpani et al., 1991). When developed, antibodies against SA have not been tested in plants (Bennett et al., 1987).

Mechanism of action

SA is a lipophilic substance, which makes it miscible with sebaceous gland and epidermal lipids of hair follicles and skin, although exact mechanism of keratolytic and comedolytic effects of salicylic acid (SA) is unclear (Arif, 2015). Salicylic acid on acne patient reduces the production of the sebum showing its therapeutic effect. Marczyk and his colleagues applied 30% SA peels and 50% pyruvic acid on acne vulgaris patients and noted that the sebumetric effect of salicylic acid stronger than pyruvic acid (Marczyk et al., 2014). SA decreases the adhesion of corneocytes, this leads to relaxation of the cell and eventually detachment (Roberts et al., 1980)(Davies & Marks, 1976). Although SA is a lipophilic substance, it removes the intercellular lipids which is stronger adhesion than cornified envelope that surrounds the epithelial cells (Lazo et al., 1995). Many dermatologists have chemically peeled the skin using antihyperplastic effects of the Salicylic acid (Swinehart, 1992)(Aronsohn, 1984).

Keratolytic/Desmolytic effect

Many genetic and epigenetic factors can cause the development of the psoriasis. Molecular researches and clinical experiences have indicated that the malfunction of epidermal permeability barrier can directly support the underlying pathophysiology of psoriasis (Ye et al., 2014)(Bergboer et al., 2012). Clinical psoriasis can be reduced by using occlusive agent, it can treat the functioning of epidermal permeability barrier (Friedman, 1987)(Fluhr et al., 2008).

Keratolytic/desmolytic medicines and moisturizers or barrier-repair formulations are effective topical treatment are observed in clinical researches (Draelos, 2008) (Elias & MENON, 1991). Predominant lipids of ceramides stratum corneum (SC), contributes to intercellular lipids bilayer, it directly helps to control transdermal water loss (TEWL) (Elias & MENON, 1991)(Del Rosso & Levin, 2011). Stratum corneum is the upper layer of the skin have ceramides and it maintains the moisture of the skin and strength of skin protective barrier. Quantity of ceramides decides the strength and quality of the skin barrier. This causes dryness and skin damage. Reduced ceramides levels found in the psoriasis patient and they have dry skin, this dryness is closely related to psoriasis (Imokawa et al., 1991)(Chung et al., 2002). The abnormal cell growth of the epidermis layer of skin can be caused by decreased level of ceramides. Decreasing level of ceramides not only

causes dryness, it also affects the moisture and barrier function of the skin, reduced ceramide is related to multiple skin problems (Lew et al., 2006).

Limitation and Challenges in psoriasis

Patients who switched to other biologic have improvement as the new patient, switching to the other biologic is better option to control or improve psoriasis condition, in case initial biologic is inefficient or have side effect (Ormerod, 2010)(Shin et al., 2023). In USA, one study was

conducted on psoriasis patients who were being treated with biologic medicines. It was observed that 14.4% patient changed their biologic drug in 1 year and it became 26.0% in 2 years, in most of the cases patients chose IL-23 inhibitor because of its lower risk and long-term effect as compare to TNF inhibitor, IL-17 inhibitor, IL -12/23 inhibitor. Factors that can help predict the drug change are age, female gender, patients familiar to the other immune drugs (Armstrong et al., 2023).

High cost is the significant problem of the biologic therapy, this enhances the financial burden and load of healthcare system. According to a survey psoriasis patients want affordable and cost effective treatments (Porter et al., 2022). But, as biologic drugs patients reach to expiration, development of biosimilar started these are similar as biologic drug and economic, cost-effective, accessible. The biosimilars of TNF-α inhibitor are effective as the original drug (Egeberg et al., 2018)(García-Beloso et al., 2022). Similarly, it was expected that IL-17 and IL-23 inhibitor provide the similar effect as the original drug (Lee & Kim, 2023). Also, some psoriasis subtypes, like nail (Pasch, 2016), scalp (Ghafoor et al., 2022) and genital (Kelly & Ryan, 2019) , palmoplantar psoriasis (PPP) (Misiak-Galazka et al., 2020), generalized pustular psoriasis (GPP) (Hoegler et al., 2018) Contains different type of problem and challenges in order to management and treatment of the psoriasis. currently available treatments of the psoriasis subtypes often fail in providing satisfactory results. There is more need of researches and development for the effective treatment and tailored approach to improve the current treatment available for the treatment of these subtype psoriasis patients (Lee & Kim, 2023).

Salicylic acid for the psoriasis management

3% aqueous salicylic acid which has lipophilic nature with turmeric gel and regular use of dry shea butter with moisturizer containing salicylic (Draelos, 2022). This treatment strategy combines anti-inflammatory effect, moisturization, exfoliation, antioxidants effect this formulation is effective and pleasant for the psoriasis treatment (Sarafian et al., 2015)(Sun et al., 2013).

Salicylic acid can effectively help in psoriasis management (Elmets et al., 2021). Anti-inflammatory and exfoliating properties of the salicylic acid makes it highly effective and reduces lessens scale, induration, irritation, erythema (Torsekar & Gautam, 2017). Additionally, salicylic acid contains antiproliferative property and improves treatment effect onset of action by enhancing the penetration rate (Draelos, 2022).

Salicylic acid encourages stratum corneum desquamation, as a keratolytic agent (Torsekar & Gautam, 2017). This agent breaks keratin of the skin and creates water-binding sites, and helps to desquamate the scaly layer of the skin (Draelos, 2022). Salicylic acid is being used for the skin treatment, since 1950s in combination with other medicines, including topical corticosteroids or on its own (Chandra et al., 2019)(CIMITAN & PACCHIANI, 1951).exfoliating moisturizers containing occlusive agents (petrolatum, shea buffer, and paraffin) with 3% salicylic acid gel (which can damage the skin barrier and cause irritation) for skin hydration, barrier repair and protection (Purnamawati et al., 2017)(Ghadially et al., 1992). The irritating effects of salicylic acid gel can be decreased by applying moisturizer on the salicylic acid gel, this moisturizer locks moisture and create protective layer and creates ideal environment for skin barrier repair (Draelos, 2022). Salicylic acid with dithranol cream was applied and rinsed/removed after certain time period to understand its role, 2% salicylic acid with dithranol (D+S) one side and only dithranol (D) on the other side of the body for comparison on outpatient treatment basis. Patient was observed for 3 times a week until the optimal dithranol concentration was found for the patient; after the concentration was found, patient was scheduled to visit 1 time a week (de Mare et al., 1989). Psoriasis area severity index (PASI) is used to measure clinical effectiveness (van Erp et al., 1989)(de Mare et al., 1990).

Using salicylic acid alone can only provide softening and dead skin cleaning for the upper layer of the skin (Pharmacists., 1984). Salicylic acid is usually combined with other drugs such as corticosteroids , to increase its effect (Lebwohl, 1999).

Salicylic acid and corticosteroids combination on psoriasis

The two properties such as anti-inflammatory and penetration rate of the corticosteroids decides its effectiveness in the treatment of the psoriasis (Lebwohl, 1999).

The salicylic acid's dead skin exfoliating effect is same as the occlusion, using combination of corticosteroid and salicylic acid 2% improve and enhances the corticosteroids penetration rate and this was observed in a study (Krochmal et al., 1989).

40 patients having moderate to severe psoriasis, seborrheic dermatitis or neurodermatitis of scalp, were treated with alcohol-based lotion for 21 days in double-blind study. This lotion contains 0.05% betamethasone-17, 21-dipropionate with 2% salicylic acid, or placebo. It was observed that salicylic acid enhances the effectiveness of the corticosteroids and it has combined effect in the erythema, pruritus both. These observation proves that combination of salicylic acid and corticosteroids enhances the activity of corticosteroids (Elie et al., 1983).

2. Problem Identification

1. Psoriasis is a chronic disease and various factors are involved in its development such as environment, genetics, and other condition, there are different type of psoriasis according to its site of occurrence.

2. Different site of occurrence makes it harder to treat such as scalp, nail, hand palm or foot sole so topical treatment should be designed according to the site of application for better, safe and highly effective for psoriasis management.

3. Salicylic acid can be used in the topical treatment for the psoriasis management but its effectiveness might be different in different patients, long term use can cause skin damage, it can show side effect in sensitive skin.

4. The concentration of salicylic acid might be different as per the site of application such as face, hands palms, foot sole, genital area etc. to enhance penetration rate and effectiveness.

5. Improper handling of salicylic acid combination can cause side effects such as dryness, itching, burning, skin irritation etc.

6. Salicylic acid alone cannot help in the psoriasis management, the combination drug that is stable with salicylic acid and enhance each other activity is necessary such as steroids, coal tar, turmeric etc.

3. Objective

1. The understand role of salicylic acid in the psoriasis management and treatment, with different formulations, combinations and forms.

2. The effect of immune (innate, adaptive) and antimicrobial defense system psoriasis development in the psoriasis development, and the principle of innate defense system and receptors of antimicrobial defense, immune system.

3. Understand the epidemiology of the psoriasis, and the criteria's that promotes psoriasis development.

4. Methodology

For the research of this article the information was collected from different articles obtained from:

- Google Scholar,
- ResearchGate,
- NIH (national institute of health),
- PubMed etc. peer-viewed and reliable scientific databases.

Search strategy Inclusion:

- Psoriasis- related studies,
- Salicylic acid mechanism,
- Formulation & treatment studies,
- Genetic and environmental factor.

Exclusion:

- Irrelevant topics,
- Duplicate studies,
- Non- scientific sources

Articles included in the review were published between 2000-2026,the search strategy included topics such as psoriasis and its morphology, pathophysiology, diagnosis, treatment and salicylic acid mechanism of action, its keratolytic and other effect etc.

Articles that were highly focused on the formulation, techniques of treatment, combination techniques containing salicylic acid, genetic characteristics and conditions that influence the psoriasis growth.

The irrelevant subjects to the review were excluded. Studies from various geographical regions/areas are added to understand the role of climate and ecological conditions that influence psoriasis development and maintenances. Selected studies were critically analyzed and compiled for the review.

5. Result & Discussion

This review clarifies that psoriasis is an immune mediated, complex, multifactorial skin disease where genetic, immune system and environmental factors contributes in the psoriasis development. Researches has shown that keratinocyte hyperproliferation and inflammatory cytokines (TNF- α , IL-17, IL-23) promotes the psoriasis development.

Globally (worldwide) 2-3% of the population suffers with psoriasis and it eventually affects the lifestyle, psychological state and associated/linked conditions and this observation is obtained from the data of epidemiology. Multiple clinical trials and articles proves that the psoriasis condition is highly influenced by atmosphere, physical and mental health, but daily exposure of sunlight or UV exposure helps in the management of the psoriasis by improving psoriasis condition and characteristics.

Studies have shown that that salicylic acid has better result in the psoriasis management because of its properties. It also resists the lesion formation, proliferation and promotes other drugs for better effect and penetration.

Result- Salicylic acid can be used as a primary drug for the psoriasis management but this drug can be used in a combination with other drug for better and effective result, it can be used as a supportive agent.

6. Conclusion

Psoriasis is an immune-mediated inflammatory chronic skin disease that affects 2-3% population worldwide. There are multiple conditions such as genetic, atmosphere, life style etc. that promotes the development and growth of psoriasis.

Psoriasis has many forms and each form is named according to its appearance and site of occurrence such as nail psoriasis, scalp psoriasis, plaque psoriasis etc. and there are higher cases of plaque psoriasis.

Salicylic acid and its properties help to increase the bioavailability of the other drugs that are used in combination with it and reduces dead cell accumulation. When the combination of salicylic acid and corticosteroids used it was observed that they were highly effective on the other comorbid issues with psoriasis, but sometimes the action of these drugs can be resisted in some patients.

In conclusion- psoriasis cannot be eliminated permanently but it can be reduced with the help of personalized treatment, regular routine, regular sun exposure etc. and these options can help in the improvement of the daily life of the patients

7. Reference

1. Arif, T. (2015). Salicylic acid as a peeling agent: a comprehensive review. *Clinical, Cosmetic and Investigational Dermatology*, *8*, 455–461. https://doi.org/10.2147/CCID.S84765

Armstrong, A. W., Patel, M., Li, C., Garg, V., Mandava, M. R., & Wu, J. J. (2023). Real-world switching patterns and associated characteristics in patients with psoriasis treated with biologics in the United States. *The Journal of Dermatological Treatment*, *34*(1), 2200870. https://doi.org/10.1080/09546634.2023.2200870

Aronsohn, R. B. (1984). Hand chemosurgery. *The American Journal of Cosmetic Surgery*, *1*(3), 24–28.

Bennett, A. P., Gallacher, G., & Landon, J. (1987). The raising and characterisation of antibodies to salicylate. *Annals of Clinical Biochemistry*, *24 (Pt 4)*, 374–384. https://doi.org/10.1177/000456328702400406

Bergboer, J. G. M., Zeeuwen, P. L. J. M., & Schalkwijk, J. (2012). Genetics of psoriasis: evidence for epistatic interaction between skin barrier abnormalities and immune deviation. *Journal of Investigative Dermatology*, *132*(10), 2320–2331.

Chandra, A., Aggarwal, G., Manchanda, S., & Narula, A. (2019). Development of Topical Gel of Methotrexate Incorporated Ethosomes and Salicylic Acid for the Treatment of Psoriasis. *Pharmaceutical Nanotechnology*, *7*(5), 362–374. https://doi.org/10.2174/2211738507666190906123643

Chung, S., Kong, S., Seong, K., & Cho, Y. (2002). Gamma-linolenic acid in borage oil reverses epidermal hyperproliferation in guinea pigs. *The Journal of Nutrition*, *132*(10), 3090–3097. https://doi.org/10.1093/jn/131.10.3090

CIMITAN, O., & PACCHIANI, G. (1951). [Parenteral administration of associated salicylic acid-resorcinol and antipyrine in the treatment of psoriasis]. *Giornale di scienze mediche*, *6*(5), 93–96.

Davies, M., & Marks, R. (1976). Studies on the effect of salicylic acid on normal skin. *The British Journal of Dermatology*, *95*(2), 187–192. https://doi.org/10.1111/j.1365-2133.1976.tb00825.x

de Mare, S., Calis, N., den Hartog, G., & van de Kerkhof, P. C. (1989). Outpatient treatment with short-contact dithranol. The impact of frequent concentration adjustments. *Acta Dermato-Venereologica*,

69(5), 449–451.

de Mare, S., van Erp, P. E., Ramaekers, F. C., & van de Kerkhof, P. C. (1990). Flow cytometric quantification of human epidermal cells expressing keratin 16 in vivo after standardized trauma. *Archives of Dermatological Research*, *282*(2), 126–130. https://doi.org/10.1007/BF00493471

Del Rosso, J. Q., & Levin, J. (2011). The clinical relevance of maintaining the functional integrity of the stratum corneum in both healthy and disease-affected skin. *The Journal of Clinical and Aesthetic Dermatology*, *4*(9), 22.

Draelos, Z. D. (2008). Moisturizing cream ameliorates dryness and desquamation in participants not receiving topical psoriasis treatment. *Cutis*, *82*(3), 211–216.

Draelos, Z. D. (2022). The Efficacy and Tolerability of Turmeric and Salicylic Acid in Psoriasis Treatment. *Psoriasis: Targets and Therapy*, *12*, 63–71. https://doi.org/10.2147/PTT.S360448

Egeberg, A., Ottosen, M. B., Gniadecki, R., Broesby-Olsen, S., Dam, T. N., Bryld, L. E., Rasmussen, M. K., & Skov, L. (2018). Safety, efficacy and drug survival of biologics and biosimilars for moderate-to-severe plaque psoriasis. *The British Journal of Dermatology*, *178*(2), 509–519. https://doi.org/10.1111/bjd.16102

Elias, P. M., & MENON, G. K. (1991). Structural and lipid biochemical correlates of the epidermal permeability barrier. *Advances in Lipid Research*, *24*, 1–26.

Elie, R., Durocher, L. P., & Kavalec, E. C. (1983). Effect of salicylic acid on the activity of betamethasone-17,21-dipropionate in the treatment of erythematous squamous dermatoses. *The Journal of International Medical Research*, *11*(2), 108–112. https://doi.org/10.1177/030006058301100208

Elmets, C. A., Korman, N. J., Prater, E. F., Wong, E. B., Rupani, R. N., Kivelevitch, D., Armstrong, A. W., Connor, C., Cordoro, K. M., Davis, D. M. R., Elewski, B. E., Gelfand, J. M., Gordon, K. B., Gottlieb, A. B., Kaplan, D. H., Kavanaugh, A., Kiselica, M., Kroshinsky, D., Lebwohl, M., … Menter, A. (2021). Joint AAD-NPF Guidelines of care for the management and treatment of psoriasis with topical therapy and alternative medicine modalities for psoriasis severity measures. *Journal of the American Academy of Dermatology*, *84*(2), 432–470. https://doi.org/10.1016/j.jaad.2020.07.087

Fluhr, J. W., Cavallotti, C., & Berardesca, E. (2008). Emollients, moisturizers, and keratolytic agents in

psoriasis. *Clinics in Dermatology*, *26*(4), 380–386.

Friedman, S. J. (1987). Management of psoriasis vulgaris with a hydrocolloid occlusive dressing. *Archives of Dermatology*, *123*(8), 1046–1052.

García-Beloso, N., Altabás-González, I., Samartín-Ucha, M., Gayoso-Rey, M., De Castro-Parga, M. L., Salgado-Barreira, Á., Cibeira-Badia, A., Piñeiro-Corrales, M. G., González-Vilas, D., Pego-Reigosa, J. M., & Martínez-López de Castro, N. (2022). Switching between reference adalimumab and biosimilars in chronic immune-mediated inflammatory diseases: A systematic literature review. *British Journal of Clinical Pharmacology*, *88*(4), 1529–1550. https://doi.org/10.1111/bcp.15101

George, Y., Aldeen, M., & Garnavi, R. (2017). A Pixel-Based Skin Segmentation in Psoriasis Images Using Committee of Machine Learning Classifiers. *2017 International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, 1–8. https://doi.org/10.1109/DICTA.2017.8227398

Ghadially, R., Halkier-Sorensen, L., & Elias, P. M. (1992). Effects of petrolatum on stratum corneum structure and function. *Journal of the American Academy of Dermatology*, *26*(3 Pt 2), 387–396. https://doi.org/10.1016/0190-9622(92)70060-s

Ghafoor, R., Patil, A., Yamauchi, P., Weinberg, J., Kircik, L., Grabbe, S., & Goldust, M. (2022). Treatment of Scalp Psoriasis. *Journal of Drugs in Dermatology : JDD*, *21*(8), 833–837. https://doi.org/10.36849/JDD.6498

Hoegler, K. M., John, A. M., Handler, M. Z., & Schwartz, R. A. (2018). Generalized pustular psoriasis: a review and update on treatment. *Journal of the European Academy of Dermatology and Venereology : JEADV*, *32*(10), 1645–1651. https://doi.org/10.1111/jdv.14949

Imokawa, G., Abe, A., Jin, K., Higaki, Y., Kawashima, M., & Hidano, A. (1991). Decreased level of ceramides in stratum corneum of atopic dermatitis: an etiologic factor in atopic dry skin? *Journal of Investigative Dermatology*, *96*(4), 523–526.

Janzen, D. H. (1981). Secondary Plant Products, Edited by E. A. Bell & B. V. Charlwood. (Encyclopedia of Plant Physiology, New Series, Vol. 8.) Springer-Verlag, Berlin—Heidelberg—New York: xvi + 674 pp., illustr., 23 × 16 × 3.5 cm, DM. 198 (approx. US$ 110.90), 1980. *Environmental Conservation*, *8*(1), 79–79. https://doi.org/10.1017/S0376892900026904

Kasolang, S. B., Adlina, W. A., Rahman, N. A., & Roseley, N. R. N. (2020). *Common skin disorders: A review*. https://api.semanticscholar.org/CorpusID:236099823

Kelly, A., & Ryan, C. (2019). Genital Psoriasis: Impact on Quality of Life and Treatment Options. *American Journal of Clinical Dermatology*, *20*(5), 639–646. https://doi.org/10.1007/s40257-019-00447-5

Kim, W. B., Jerome, D., & Yeung, J. (2017). Diagnosis and management of psoriasis. *Canadian Family Physician Medecin de Famille Canadien*, *63*(4), 278–285.

Krochmal, L., Wang, J. C., Patel, B., & Rodgers, J. (1989). Topical corticosteroid compounding: effects on physicochemical stability and skin penetration rate. *Journal of the American Academy of Dermatology*, *21*(5 Pt 1), 979–984. https://doi.org/10.1016/s0190-9622(89)70286-3

Lazo, N. D., Meine, J. G., & Downing, D. T. (1995). Lipids are covalently attached to rigid corneocyte protein envelopes existing predominantly as beta-sheets: a solid-state nuclear magnetic resonance study. *The Journal of Investigative Dermatology*, *105*(2), 296–300. https://doi.org/10.1111/1523-1747.ep12318985

Lebwohl, M. (1999). The role of salicylic acid in the treatment of psoriasis. *International Journal of Dermatology*, *38*(1).

Lee, H.-J., & Kim, M. (2023). Challenges and Future Trends in the Treatment of Psoriasis. *International Journal of Molecular Sciences*, *24*(17). https://doi.org/10.3390/ijms241713313

Lew, B.-L., Cho, Y., Kim, J., Sim, W.-Y., & Kim, N.-I. (2006). Ceramides and cell signaling molecules in psoriatic epidermis: reduced levels of ceramides, PKC-alpha, and JNK. *Journal of Korean Medical Science*, *21*(1), 95–99. https://doi.org/10.3346/jkms.2006.21.1.95

Marczyk, B., Mucha, P., Budzisz, E., & Rotsztejn, H. (2014). Comparative study of the effect of 50% pyruvic and 30% salicylic peels on the skin lipid film in patients with acne vulgaris. *Journal of Cosmetic Dermatology*, *13*(1), 15–21. https://doi.org/10.1111/jocd.12050

Michalek, I., Loring, B., & John, S. (2016). *WHO Global report on psoriasis*.

Misiak-Galazka, M., Zozula, J., & Rudnicka, L. (2020). Palmoplantar Pustulosis: Recent Advances in Etiopathogenesis and Emerging Treatments. *American Journal of Clinical Dermatology*, *21*(3), 355–370. https://doi.org/10.1007/s40257-020-00503-5

Mitchell, A. G., & Broadhead, J. F. (1967). Hydrolysis of solubilized aspirin. *Journal of Pharmaceutical Sciences*, *56*(10), 1261–1266. https://doi.org/10.1002/jps.2600561009

Ormerod, A. D. (2010). Switching biologics for psoriasis. In *The British journal of dermatology* (Vol. 163, Issue 4, pp. 667–669). https://doi.org/10.1111/j.1365-2133.2010.09999.x

Papp, K., Gulliver, W., Lynde, C., Poulin, Y., & Ashkenas, J. (2011). Canadian guidelines for the management of plaque psoriasis: overview. *Journal of Cutaneous Medicine and Surgery*, *15*(4), 210–219. https://doi.org/10.2310/7750.2011.10066

Pasch, M. C. (2016). Nail Psoriasis: A Review of Treatment Options. *Drugs*, *76*(6), 675–705. https://doi.org/10.1007/s40265-016-0564-5

Pharmacists., A. S. of H. (1984). American hospital formulary service drug information. *American Hospital Formulary Service Drug Information.*

Porter, C., Woods, A., Mendelow, M., Purvis, C., & Feldman, S. (2022). Unmet Needs in Psoriasis Patients. *Journal of Drugs in Dermatology : JDD*, *21*(8), 839–844. https://doi.org/10.36849/JDD.6589

Purnamawati, S., Indrastuti, N., Danarti, R., & Saefudin, T. (2017). The Role of Moisturizers in Addressing Various Kinds of Dermatitis: A Review. *Clinical Medicine & Research*, *15*(3–4), 75–87. https://doi.org/10.3121/cmr.2017.1363

Raskin, I. (1992). Salicylate, A New Plant Hormone 1. *Plant Physiology*, *99*(3), 799–803. https://doi.org/10.1104/pp.99.3.799

Raskin, I., Ehmann, A., Melander, W. R., & Meeuse, B. J. (1987). Salicylic Acid: a natural inducer of heat production in arum lilies. *Science (New York, N.Y.)*, *237*(4822), 1601–1602. https://doi.org/10.1126/science.237.4822.1601

RASKIN, I., Skubatz, H., Tang, W., & Meeuse, B. J. D. (1990). Salicylic Acid Levels in Thermogenic and Non-Thermogenic Plants. *Annals of Botany*, *66*, 369–373. https://doi.org/10.1093/oxfordjournals.aob.a088037

Raskin, I., Turner, I. M., & Melander, W. R. (1989). Regulation of heat production in the inflorescences of an Arum lily by endogenous salicylic acid. *Proceedings of the National Academy of Sciences of the United States of America*, *86*(7), 2214–2218. https://doi.org/10.1073/pnas.86.7.2214

Roberts, D. L., Marshall, R., & Marks, R. (1980). Detection of the action of salicylic acid on the normal stratum corneum. *The British Journal of Dermatology*, *103*(2), 191–196. https://doi.org/10.1111/j.1365-2133.1980.tb06590.x

Sarafian, G., Afshar, M., Mansouri, P., Asgarpanah, J., Raoufinejad, K., & Rajabi, M. (2015). Topical Turmeric Microemulgel in the Management of Plaque Psoriasis; A Clinical Evaluation. *Iranian Journal of Pharmaceutical Research : IJPR*, *14*(3), 865–876.

Shin, J.-O., Shin, B. S., Bae, K.-N., Shin, K., Kim, H.-S., Ko, H.-C., Kim, M.-B., & Kim, B. (2023). Review of the reasons for and effectiveness of switching biologics for psoriasis treatment in Korea. *Indian Journal of Dermatology, Venereology and Leprology*, *89*(6), 928. https://doi.org/10.25259/IJDVL_308_2022

Sun, J., Zhao, Y., & Hu, J. (2013). Curcumin inhibits imiquimod-induced psoriasis-like inflammation by inhibiting IL-1beta and IL-6 production in mice. *PloS One*, *8*(6), e67078.

Swinehart, J. M. (1992). Salicylic acid ointment peeling of the hands and forearms. Effective nonsurgical removal of pigmented lesions and actinic damage. *The Journal of Dermatologic Surgery and Oncology*, *18*(6), 495–498. https://doi.org/10.1111/j.1524-4725.1992.tb03311.x

Torsekar, R., & Gautam, M. M. (2017). Topical therapies in psoriasis. *Indian Dermatology Online Journal*, *8*(4), 235–245.

van Erp, P. E., Rijzewijk, J. J., Boezeman, J. B., Leenders, J., de Mare, S., Schalkwijk, J., van de Kerkhof, P. C., Ramaekers, F. C., & Bauer, F. W. (1989). Flow cytometric analysis of epidermal subpopulations from normal and psoriatic skin using monoclonal antibodies against intermediate filaments. *The American Journal of Pathology*, *135*(5), 865–870.

Yalpani, N., Silverman, P., Wilson, T. M., Kleier, D. A., & Raskin, I. (1991). Salicylic acid is a systemic signal and an inducer of pathogenesis-related proteins in virus-infected tobacco. *The Plant Cell*, *3*(8), 809–818. https://doi.org/10.1105/tpc.3.8.809

Ye, L., Lv, C., Man, G., Song, S., Elias, P. M., & Man, M.-Q. (2014). Abnormal epidermal barrier recovery in uninvolved skin supports the notion of an epidermal pathogenesis of psoriasis. *The Journal of Investigative Dermatology*, *134*(11), 2843.